

Exponent Simplification Exercises — Answer Sheet

1 1 Basic Answers

1. x^8

2. a^6

3. m^{12}

4. $16y^2$

5. 1

6. $\frac{1}{k^5}$

7. $27a^3b^3$

8. $\frac{1}{p^7}$

2 2 Intermediate Answers

1. $\frac{1}{x^2}$

2. c^2

3. $\frac{25}{t^4}$

4. $\frac{1}{x^3y^6}$

5. $\frac{1}{a^3}$

6. $\frac{1}{s}$

7. $\frac{9n^2}{m^3}$

8. $\frac{1}{z^6}$

3 3 Advanced Answers

1. $\frac{2}{a^2b}$
2. $\frac{9}{x^5y^6}$
3. $\frac{1}{125}$
4. $\frac{4q^2}{p^3}$
5. $\frac{1}{m^3}$

4 4 Evaluation Answers

1. $a^{m+n} = a^m \cdot a^n = 5 \times 3 = 15$
2. $a^{3m-n} = \frac{(a^m)^3}{a^n} = \frac{5^3}{3} = \frac{125}{3}$